Ali Pourmand

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https://github.com/AliPourmand/
https://alipourmand.github.io/
https://scholar.google.com/citations?user=jDj2Mq8AAAAJ&hl=en

Research Interests

- Keywords: Binary Stellar Evolution, Second Generation Planet Formation, Pulsar Planets, Post-AGB Disks
- Main tools: Hydrodynamical Simulations, Stellar Evolution Codes, Analytical Modelling.

Education

2025 – 2028 **Ph.D., Macquarie University** Astrophysics.
Thesis title: *Investigating Second Generation Planet Formation*

2020 – 2023 **M.Sc., University of Alberta** Physics.
Thesis title: *Properties of Binary Systems in One-Dimensional Approximations and Applications*. Link to thesis here

2018 – 2020 **Minor's Degree, University of Tehran** Physics.

2015 – 2020 **B.Sc., University of Tehran** Mechanical Engineering.
Thesis title: *Geometric design of a solar dish for use in the Stirling cycle*

Publications

Preprints

1. **Pourmand, A.**, Kamath, D., De Marco, O., & Wardle, M. (2025). Second-generation planet formation in post-agb discs: Testing the role of gravitational instability. arXiv: 2509.03894 [astro-ph.EP]. Retrieved from https://arxiv.org/abs/2509.03894

Published

1. Aggarwal, N., **Pourmand, A.**, Shojai, F., & Parthasarathy, H. (2025). Observational constraints on chaplygin gas models in non-minimally coupled power law $f(Q)$ gravity with quasars. *Annals of Physics*, *483*, 170266. doi:10.1016/j.aop.2025.170266
2. Ivanova, N., Kundu, S., & **Pourmand, A.** (2024). Unified rapid mass transfer. *The Astrophysical Journal*. Retrieved from https://iopscience.iop.org/article/10.3847/1538-4357/ad583e
3. **Pourmand, A.**, & Ivanova, N. (2023). Properties of binary systems in a one-dimensional approximation. *The Astrophysical Journal*. Retrieved from https://iopscience.iop.org/article/10.3847/1538-4357/acd4c1/meta

Conference Papers

1. **Pourmand, A.**, & Ivanova, N. (2024). Accurate simulations of binary stars up to mass transfer ignition. In *26th meeting on research in astronomy at iasbs*. Retrieved from https://iasbs.ac.ir/~astro/26/docs/Proceeding%20(astro%2026).pdf

Talks and Poster Presentations

Academic Talks

October 2025 — **SPRIGS , University of Sydney** Investigating Second Generation Planet Formation in Post-AGB Disks

July 2025 — **Astronomical Society of Australia's Annual Science Meeting (ASM), University of Adelaide** Gravitational Instability in Post-AGB Disks: Limits on Second-Generation Planet Formation

March 2025 — **Sydney Harbour Stars and Planets, UNSW** Accurately Simulating Binary Stars in 1D

May 2024 — **26th Meeting on Research in Astronomy IASBS, Institute for Advanced Studies in Basic Sciences** Accurate Simulations of Binary Stars up to Mass Transfer Ignition

April 2024 — **Department of Physics, University of Tehran** Modelling Phenomena in Modern Astrophysics: Binary Stars

Weekly Seminar, School of Astronomy, Institute for Research in Fundamental Sciences (IPM) Modelling Phenomena in Modern Astrophysics: Binary Stars

July 2022 — **OAC virtual talk, School of Astronomy, Institute for Research in Fundamental Sciences (IPM)** Current problems in binary evolution: the initial conditions for a common envelope event

Public Science Talks

2025 — **Astronomy Open Night, Macquarie University** Born of Dust and Chaos: The Story of Planet Formation

Poster Presentations

April 2025 — **2025 Physics Research Poster Presentation Network Event, Australian Institute of Physics - NSW Branch** Pourmand, A., Kamath, D., De Marco, O., Wardle, M., Van Winckel, H., Planets Born around Dying Stars

October 2022 — **GPSA Symposium, University of Alberta** Pourmand, A., Ivanova, N., Simulating Binary Systems at the Onset of the Common Envelope Phase

Awarded Telescope Times

2025 — **Co-Investigator, SPHERE** Stellar Endgames: A SPHERE Exploration of Disks and Complex Structures around Evolved Binaries, awarded time: 50h08m

Awards and Accomplishments

2025 — **International Macquarie Research Excellence Scholarship (iMQRES)** – Macquarie University, Australia.

2022 — **Student Talk Award (2nd Place)** – GPSA Symposium, University of Alberta. *Title: "The Initial Conditions of the Common Envelope Phase in Binary Stellar Evolution."*

2020 — **Graduate Research Assistant Fellowship (GRAF)** – University of Alberta, Canada.

Nationwide Graduate Entrance Exam (Philosophy of Science) – 2nd place, Sanjesh Organization, Iran

Nationwide Graduate Entrance Exam (Physics) – 28th place, Sanjesh Organization, Iran

Teaching Experience

Fall 2025 — **Casual Teacher, Macquarie University** COMP 6200; introduction to data science course.

Teaching Experience (continued)

- **Lab Demonstrator, Macquarie University** PHYS 1510; physics for engineers.
- Summer-Winter 2023 — **Tutor, Varsity Tutors** Private tutoring for high-school and college students, on various topics (mostly related to physics).
- Fall 2022 — **Teaching Assistant, University of Alberta** ASTRO 465/565, stellar astrophysics II lecture support; assisting students with installing and using the stellar evolution code MESA. Website I designed for the course here
- Winter 2021 — **Teaching Assistant, University of Alberta** Phys 124; Particles and Waves (lecture support, etc.).
- Fall 2021 — **Teaching Assistant, University of Alberta** Astro 120; Astronomy of the Solar System (Grading).
- Winter 2020 — **Teaching Assistant, University of Alberta** Phys 126 Lab; Fluids, Fields and Radiation (Lecturing and Grading).
- Summer 2018 — **Science teacher, Emam Hossein high school** Middle school level science at a summer school.

Further Research and Work Experience

Astronomy

- 2017, 2018 — **Summer Internships, School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran**, Astronomical image processing with the IRAF software. The motive was to obtain the star formation histories of several galaxies, under the supervision of Dr. Atefeh Javadi.

Engineering

- 2019 — **Bachelor's Degree Thesis, University of Tehran, Tehran, Iran**, Simulated the annual variations in the performance of solar-powered stirling systems in several cities of Iran with MATLAB.
- 2019 — **Summer internship, Sina Robotic and Medical Innovators, Tehran, Iran**, Designed components of a laparoscopic surgical instrument for robotic surgery using SOLIDWORKS, while assessing relevant medical standards for regulatory compliance.

Programming

- 2022 — **Participator, McGill Physics Hackathon, McGill University, Canada**, Modelled the ecological impact of introducing or removing species on ecosystem population dynamics, such as the effect of reintroducing wolves to Yellowstone National Park. link to project here

Workshops and Events

- 2022 — **AMUSE Summer School**, held at the Geneva Observatory, Switzerland.
- 2021 — **MESA Summer School** University of California, Santa Barbara.
- 2020 — **WestGrid Autumn School** subjects were topics such as Bash, HPC, Git, etc.

Volunteer Activities and Outreach

- 2025 — Volunteer at the Macquarie University Observatory
- Macquarie Experience Day, introduced astronomy as a career path to high school students, Macquarie University, Sydney, Australia

Volunteer Activities and Outreach (continued)

- 2022,2023 — Volunteer at the Iranian tent at Edmonton Heritage Festival (performing music, preparing food), Edmonton, Canada
- 2021 — Volunteer at the University of Alberta Observatory
- 2019 — Collaborated in organizing several ceremonies at the University of Tehran, e.g. performing several songs for the Persian New Year (Nowruz) ceremony

Skills

Programming

Languages	Fortran, Python, MATLAB, Git, Bash, Slurm
Operating Systems	Linux (Fedora, Ubunta), Windows, Mac
Markov Chain Monte Carlo	EMCEE
Other.	LaTeX

Astronomy

Stellar Evolution Code	MESA
Astronomical Catalogues	VizieR
Hydrodynamics Code	Phantom
Photometry	IRAF, DS9
Other	AMUSE

Miscellaneous

Engineering Software	SOLIDWORKS, AutoCAD, COMSOL, EES
Languages	English (native), Persian (native), French (elementary)
Music	Harmonica, Piano, Tombak (Persian instrument)
Sports	Black Belt in Kyokushin Karate, Completed a full (42km) Marathon